KP09 Choosing the Cost Function

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.8

Slides: read and download

Learning objectives

- Explain the two flaws of squared error and why a confidently wrong output cannot learn.
- Derive that with cross-entropy and a sigmoid or softmax the δ becomes $y - \hat{y}$, and explain that cross-entropy is the negative log-likelihood.
- Explain the idea and the price of discriminative learning and of Bayesian training.

KP09 Choosing the Cost Function

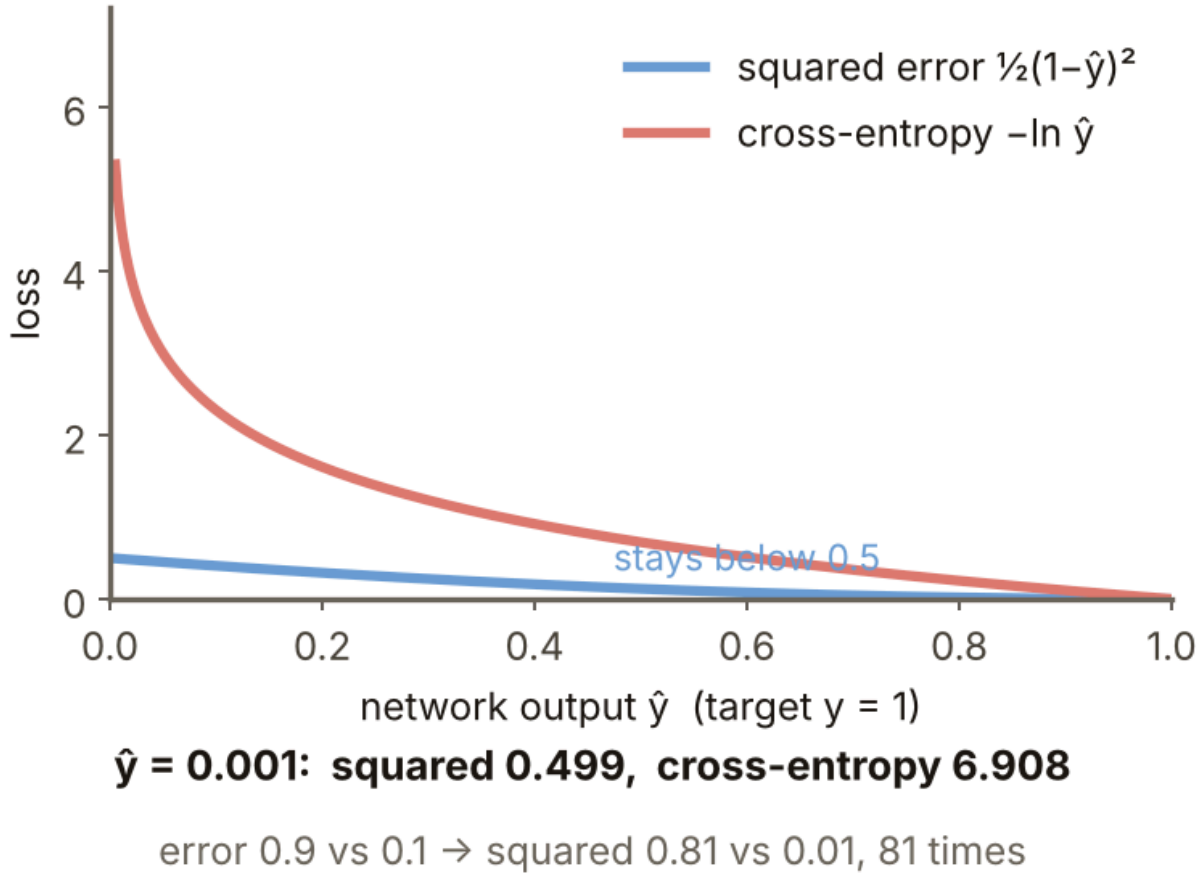
One-sentence summary: Squared error is not the only choice. For classification the cross-entropy is often the better loss: the sigmoid slope does not squeeze it away and the worse the error the harder it fixes; with softmax the backpropagation δ is simply $y - \hat{y}$.

1. Two flaws of squared error

Plain explanation: The speed-ups of KP08 all rewrite the update rule, but where the update goes depends on the cost function itself.

- **Unfair**: errors are squared before summing, so big errors dominate. Errors 0.9 and 0.1 differ 9 times, but squared they are 0.81 and 0.01, 81 times.
- **It can stall**: when the output $\hat{y}$ is wrong at an extreme (target 1, output 0.001) the squared error is only 0.499 and never grows, so gradient descent can get stuck (source citing Witt 00). Contrast: another loss, $-\ln \hat{y}$, is 6.908 at $\hat{y} = 0.001$.

Figure `lec05-kp09-loss`: both losses against the output for target 1.

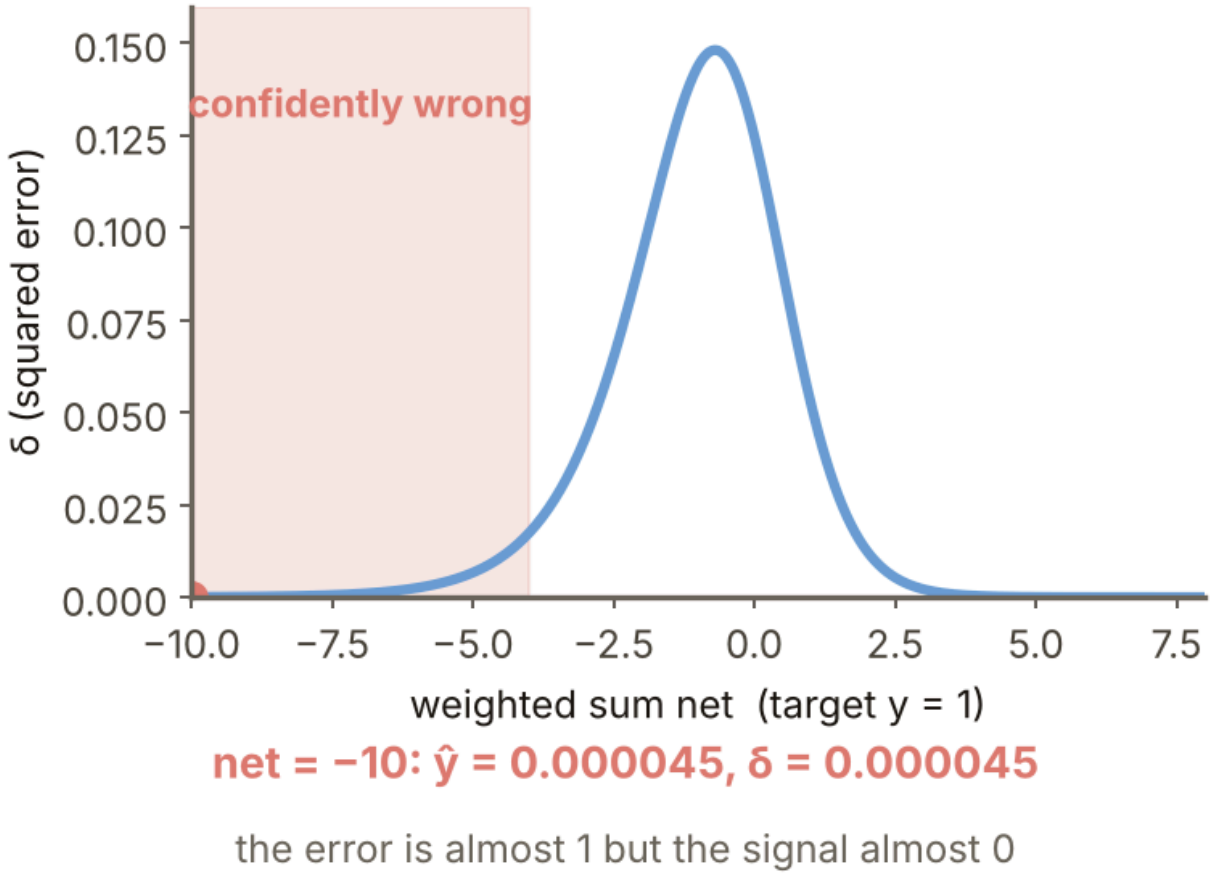


2. The plateau: δ goes to 0 when the output is wrong at an extreme

Plateau: a flat region where the cost is large but the gradient is almost 0, so gradient descent hardly moves on it.

For a single sigmoid output with squared error, $\delta = (y - \hat{y})\,\hat{y}(1 - \hat{y})$. The closer $\hat{y}$ is to 0 or 1 the smaller the slope $\hat{y}(1 - \hat{y})$, squeezing even a big error into a small signal. With target $y = 1$ and $net = -10$, $\hat{y} = 0.000045$, the error is almost 1 and $\delta = 0.000045$.

Figure `lec05-kp09-mse-delta`: the squared-error δ against the weighted sum net.



3. Cross-entropy: cancel the slope

Cross-entropy: read the output $\hat{y}$ as the probability that the target is 1. **Formal definition**:

$$J = -\big(y \ln \hat{y} + (1-y)\ln(1-\hat{y})\big),$$

which for $y = 1$ is $-\ln \hat{y}$. Derivation: $\partial J/\partial \hat{y} = -(y-\hat{y})/[\hat{y}(1-\hat{y})]$ and the sigmoid slope $\hat{y}(1-\hat{y})$ cancels it exactly, giving $\delta = y - \hat{y}$. The sign follows KP06 ($\delta = -\partial J/\partial net$); the source writes $\hat{y} - y$, which differs by a sign. At $net = -10$ the squared-error $\delta = 0.00005$ and the cross-entropy $\delta = 1.0000$.

Figure `lec05-kp09-delta-table`: $\hat{y}$ and the two δ at different net.

net	ŷ	δ squared	δ cross-ent.
−10	0.00005	0.00005	1.0000
−5	0.0067	0.00660	0.9933
−2	0.1192	0.09248	0.8808
0	0.5000	0.12500	0.5000
2	0.8808	0.01252	0.1192
5	0.9933	0.00004	0.0067

target y = 1; both δ defined as −∂J/∂net

4. Training compared

Example (designed here): one sigmoid neuron $\hat{y} = \sigma(b + wx)$, four points $x = -2, -1, 1, 2$ with targets $0, 0, 1, 1$; start $w = -3, b = 0$ so all four are wrong; $\eta = 0.5$, batch updates. Cross-entropy gets all four right in 5 epochs and squared error needs 85 (program). **Common misconception**: only cross-entropy can classify correctly. Both costs end up correct; the difference is the speed from a very wrong start.

Figure `lec05-kp09-train`: the output of the point $x = 2$ against epochs.



5. Likelihood: cross-entropy is the negative log-likelihood

Likelihood: the probability of seeing this data if the model is right; larger is better and independent samples multiply. Outputs $0.9, 0.8, 0.3, 0.1$ with targets $1, 1, 0, 0$: the probability of each target is $0.9, 0.8, 0.7, 0.9$, the product is 0.4536 and $-\ln 0.4536 = 0.7905$; the four cross-entropies $0.105 + 0.223 + 0.357 + 0.105 = 0.790$ add up to the same. Minimizing cross-entropy is maximizing likelihood.

Figure `lec05-kp09-likelihood`: probabilities and $-\ln$ of four samples.

sample	target y	output ŷ	p of the target	−ln p
1	1	0.9	0.9	0.105
2	1	0.8	0.8	0.223
3	0	0.3	0.7	0.357
4	0	0.1	0.9	0.105

product of p: 0.4536

sum of −ln p: 0.7905

−ln 0.4536 = 0.7905: the cross-entropy is the negative log-likelihood

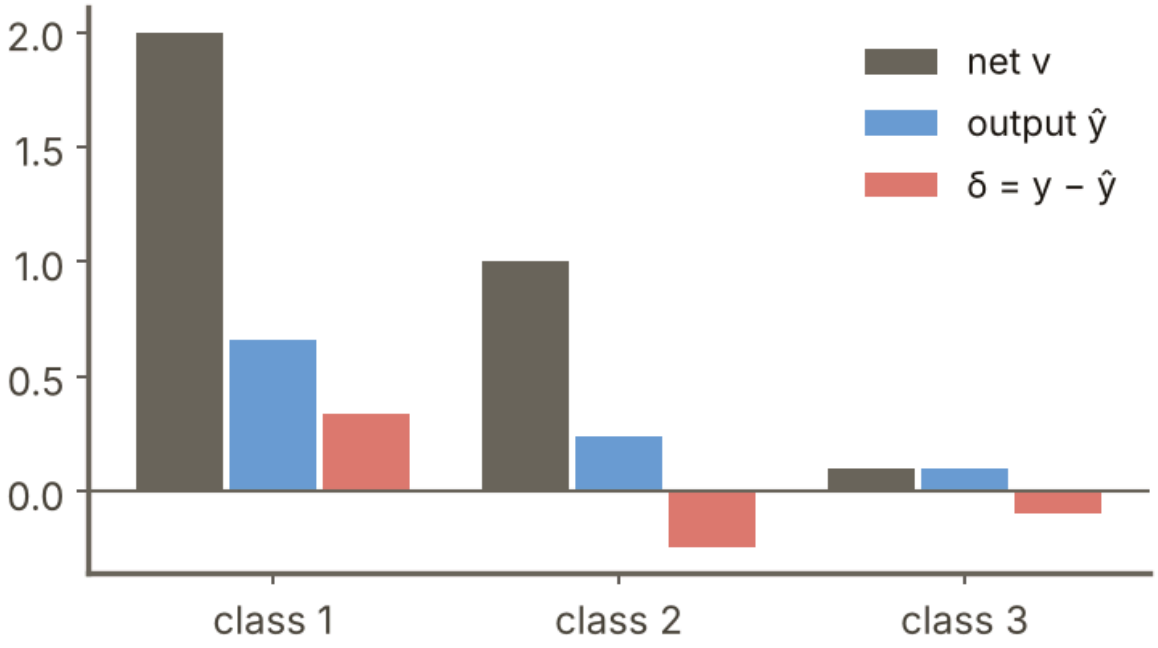
6. softmax

softmax for several classes:

$$\hat{y}_k = \frac{e^{v_k}}{\sum_{k'} e^{v_{k'}}},$$

every output lies between 0 and 1 and they sum to exactly 1. $v = (2, 1, 0.1)$ gives $\hat{y} = (0.659, 0.242, 0.099)$; with target $y = (1, 0, 0)$, $J = -\ln 0.659 = 0.417$. The $\delta_k = y_k - \hat{y}_k = (0.341, -0.242, -0.099)$ needed by backpropagation equals numerical differences (program check).

Figure `lec05-kp09-softmax`: bars of v, $\hat{y}$ and δ.



ŷ = (0.659, 0.242, 0.099), sum = 1

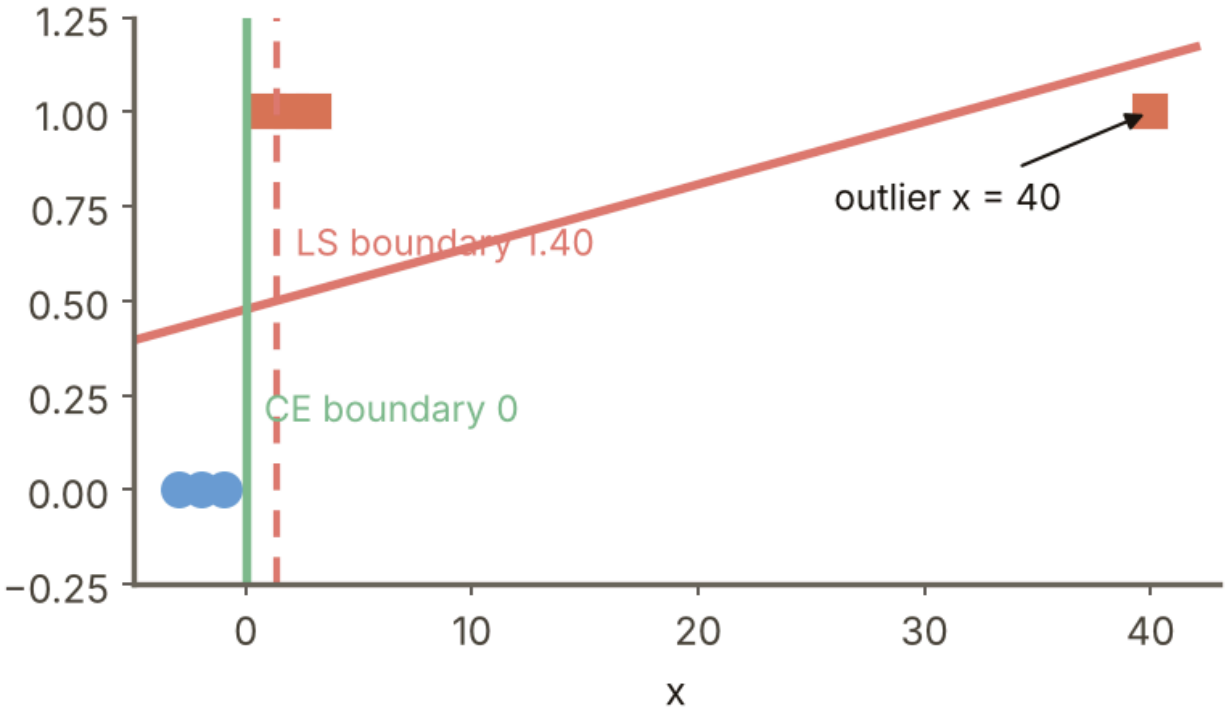
δ = (0.341, −0.242, −0.099)

checked against numerical differences of $J = -\ln \hat{y}_1$

7. Discriminative learning

Discriminative learning: since the goal is a low classification error, focus on the largest posterior and push the decision surface to lower the error directly. **Contrast example** (designed here): $x = -3, -2, -1$ are class 0; $x = 1, 2, 3$ and the outlier $x = 40$ are class 1. The squared-error line cuts at 1.40 and misclassifies $x = 1$; sigmoid with cross-entropy cuts at 0 and gets all seven right. The contrast changes both the output and the cost; the error rate itself is not differentiable, so a smooth version is used, with a risk of local minima (source, which also mentions deterministic annealing).

Figure `lec05-kp09-lsfit`: seven points and the two boundaries.



least squares line: the point x = 1 is misclassified

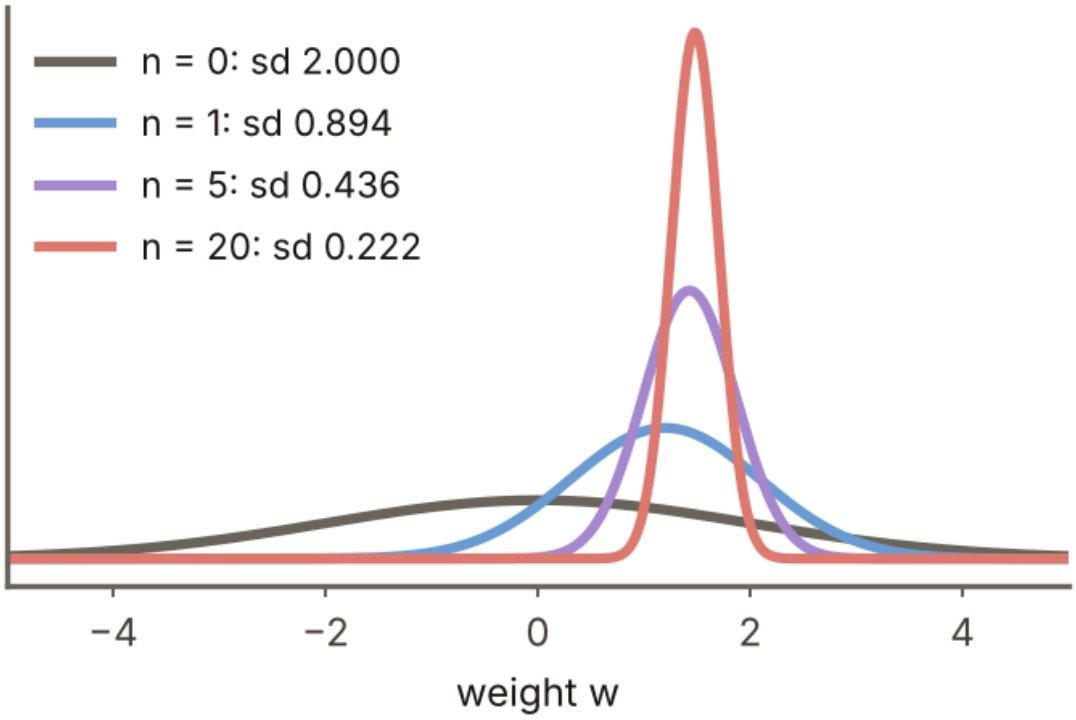
sigmoid + cross-entropy: all 7 points right

the single outlier pulled the line flat

8. Bayesian training

Bayesian training: set a prior $p(w)$ on the weights; with the data, Bayes' rule gives the posterior $p(w \mid Y) = p(Y \mid w)\, p(w)/p(Y)$; classifying averages over all w, $P(\omega_k \mid x; Y) = \int \phi_k(x; w)\, p(w \mid Y)\, dw$. **Example** (designed here): prior sd 2, every sample says $w \approx 1.5$ (noise sd 1); for $n = 0, 1, 5, 20$ the posterior sd is $2, 0.894, 0.436, 0.222$ and the mean $0, 1.2, 1.43, 1.48$. Price: a high-dimensional integral is expensive; the source reminds us that replacing least squares often gives no real gain in practice.

Figure `lec05-kp09-posterior`: the posterior for different n.



prior sd 2; every sample says w ≈ 1.5 (noise sd 1)

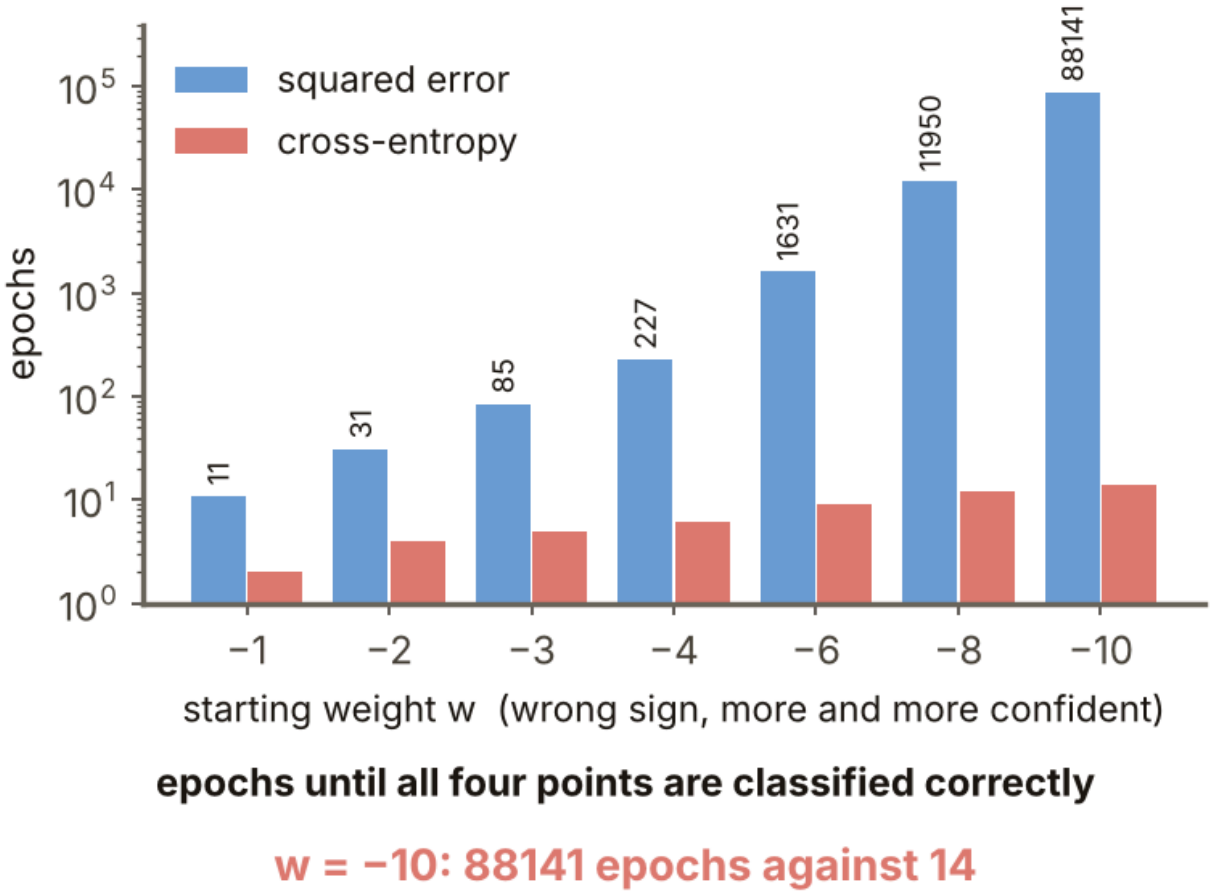
posterior mean 1.2 → 1.43 → 1.48 as n = 1, 5, 20

classifying averages over all w: a costly integral

Extension: worse starts, practice and worked solution

Worse starts: with the data of section 4 and $\eta = 0.5$, starts $w = -1, -2, -3, -4, -6, -8, -10$ need 11, 31, 85, 227, 1631, 11950 and 88141 epochs with squared error and only 2, 4, 5, 6, 9, 12 and 14 with cross-entropy. The squared-error δ shrinks exponentially as the start gets worse and the epochs grow roughly exponentially; the cross-entropy δ stays near 1.

Figure `lec05-kp09-init`: epochs needed from seven starts (log axis).



Practice: a single sigmoid output, target 1, $net = -4$.

1. What is $\hat{y} = \sigma(-4)$?
2. What are the squared-error $\delta = (1 - \hat{y})\,\hat{y}(1 - \hat{y})$ and the cross-entropy $\delta = 1 - \hat{y}$, and how many times larger is the cross-entropy one?

Figure `lec05-kp09-practice`: the setup of the practice.

one sigmoid output, target y = 1

weighted sum net = −4

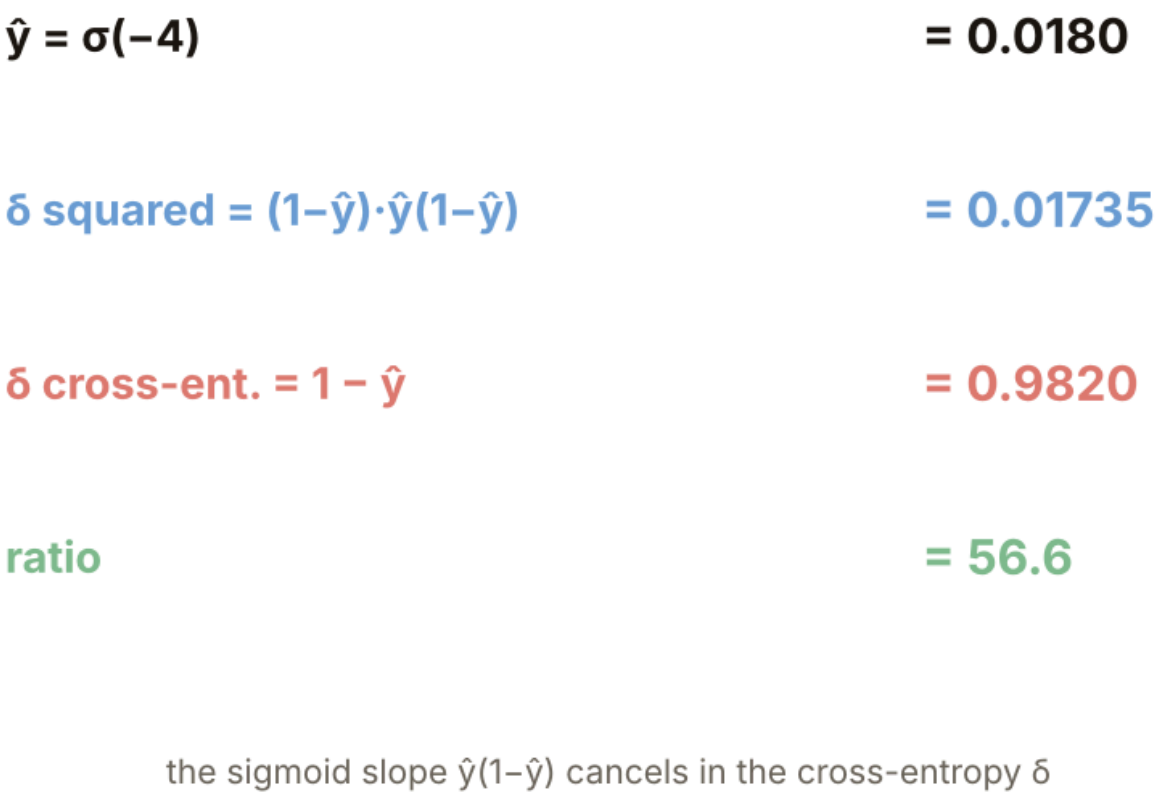
ŷ = 1 / (1 + e^4)

find δ for squared error and for cross-entropy

how many times larger is the second?

Worked solution of questions 1 and 2: (1) $\hat{y} = 1/(1+e^4) = 0.0180$. (2) Squared error $\delta = 0.9820 \times 0.0180 \times 0.9820 = 0.01735$; cross-entropy $\delta = 0.9820$; ratio $0.9820/0.01735 = 56.6$. (3) Check: the ratio is exactly $1/[\hat{y}(1-\hat{y})]$ and grows as the output gets more extreme; at $net = -10$ it is about 22000.

Figure `lec05-kp09-practice-sol`: the two δ and the ratio.



the sigmoid slope ŷ(1−ŷ) cancels in the cross-entropy δ

Terms and sources

English	中文
Plateau	高原
Cross-entropy	交叉熵
Likelihood	概似
softmax	softmax
Discriminative learning	判別式學習
Bayesian training	貝氏訓練

Source: Chapter 4 KP09, §4.8 (read 2026-10-25). The two flaws of squared error, the cross-entropy and softmax formulas, the simplified δ, and the descriptions of discriminative learning and Bayesian training follow the source; the single-neuron training comparison, the outlier contrast, the Gaussian posterior example and every

number were designed and recomputed for this lecture ([added], (program)). The sign of δ is fixed as $-\partial J / \partial net$.